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Class: T21

EXPERIMENT NO. 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios
LO Mapping: LO5
LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes in to play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

Why We Need Nagios tool?

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++,

which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).

- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also supports). When the Nagios server needs to check a resource on that host, it sends a command; NSClient++ executes the check locally and returns the result to Nagios.

Note on running Nagios on Windows

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

Implementation Steps (Native Ubuntu Localhost Monitoring):

Since you are monitoring your own Ubuntu machine natively, using a Docker container and remote agents like NSClient++ is entirely unnecessary and overly complicated. Nagios Core is built for Linux and automatically sets up monitoring for the host it is installed on (localhost) out of the box.

1. Install Nagios Core and Plugins
Open your Ubuntu terminal. We will use apt to install the native packages rather than compiling from scratch.

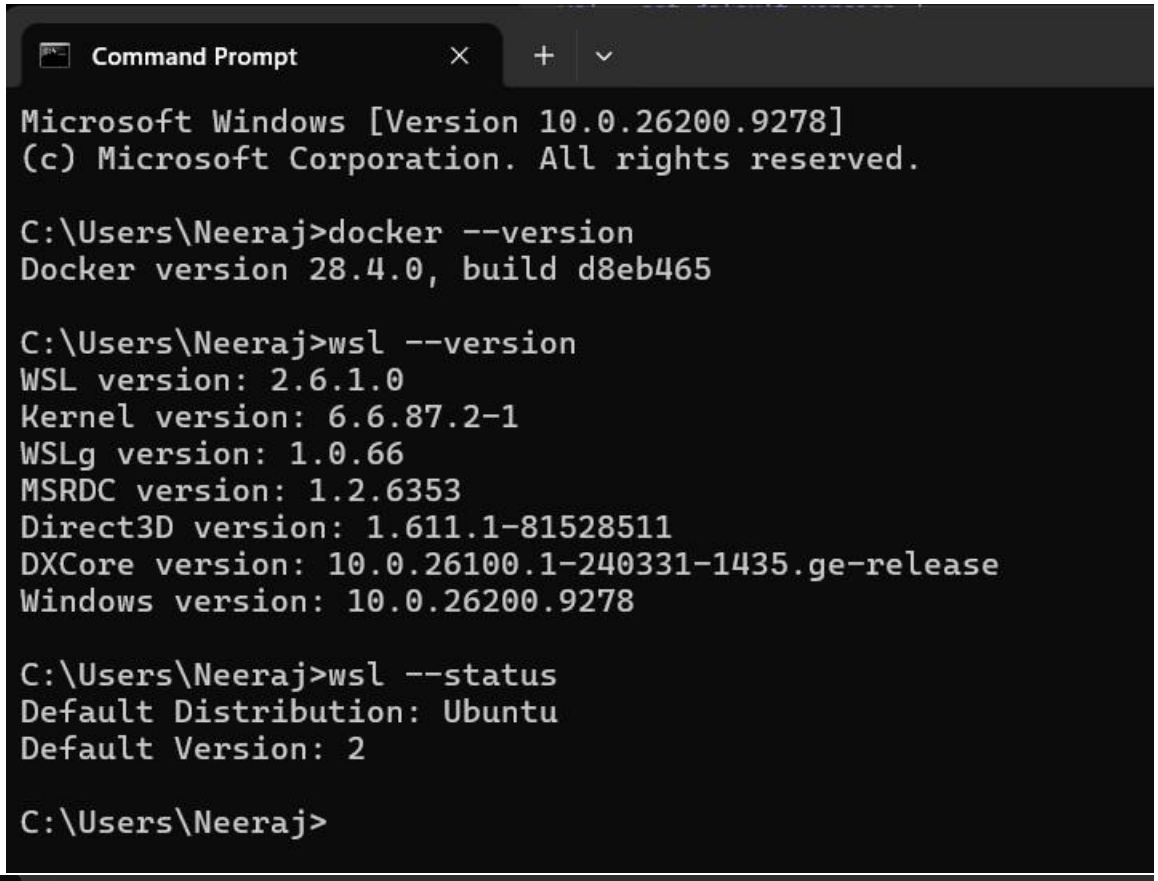
2. Set the Web Admin Password You need to set up authentication for the default nagiosadmin user to access the web interface securely.

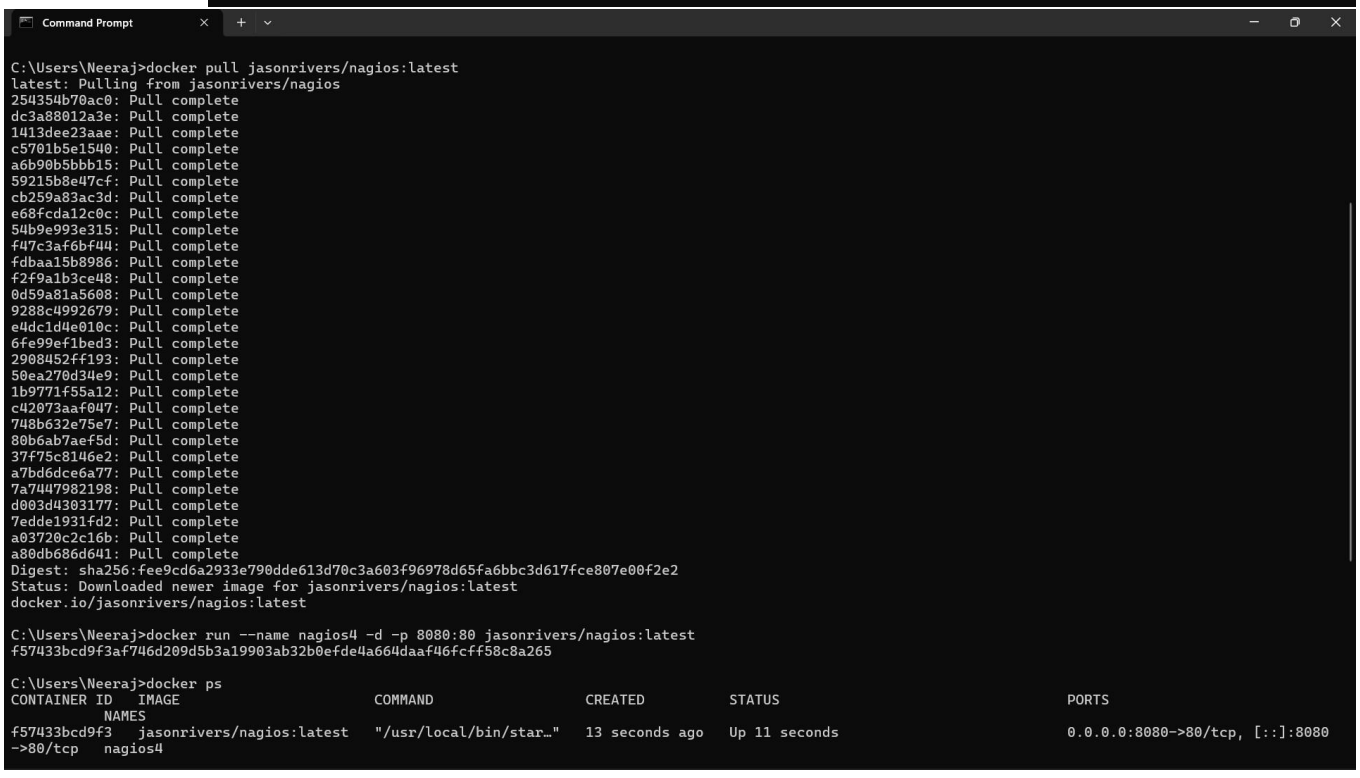
3. Restart and Enable Services
Apply the configuration changes by restarting the web server and the Nagios service.

4. **Access the Web Interface** Open a browser on your Ubuntu machine to view the dashboard.

    - **URL:** http://localhost/nagios4
    - **Username:** nagiosadmin
    - **Password:** *(The password you entered during step 2)*

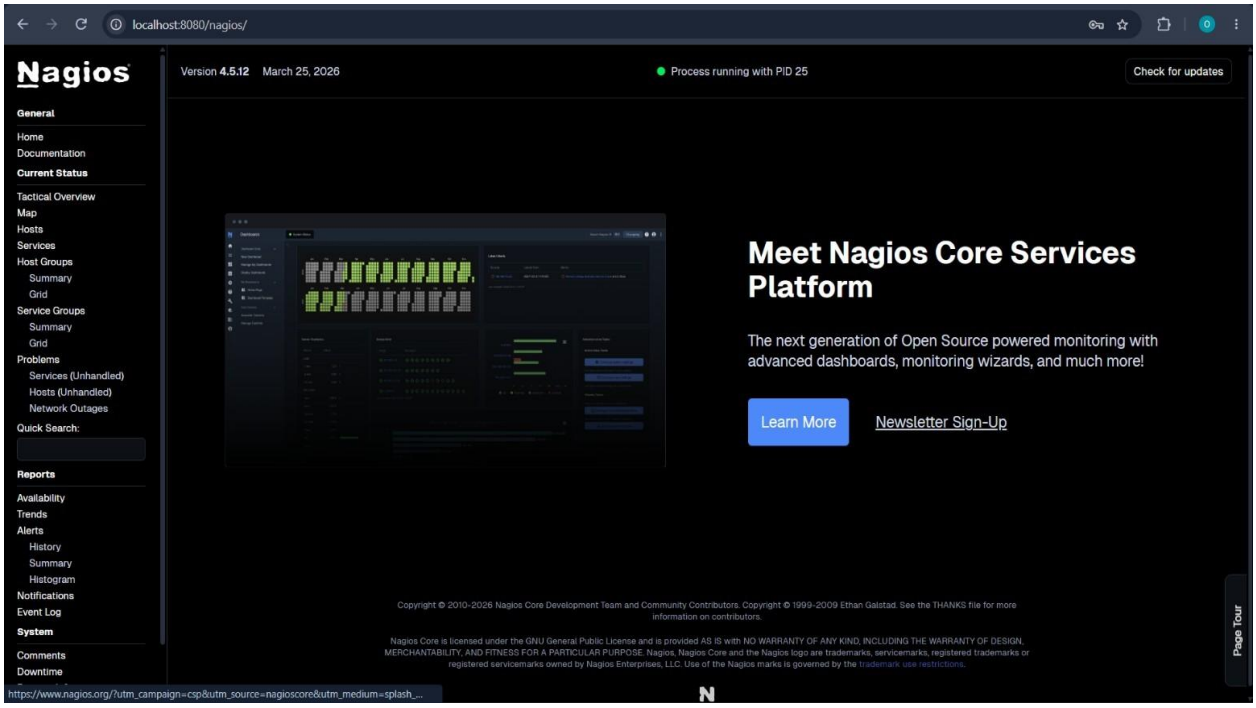
5. **Verify Localhost Monitoring** Once logged in, navigate to **Current Status -> Hosts**. Because Nagios runs natively on Linux, a configuration file for localhost is automatically active. You will see live, active monitoring for your own machine's PING, CPU load, and disk space without having to manually define the host configuration.

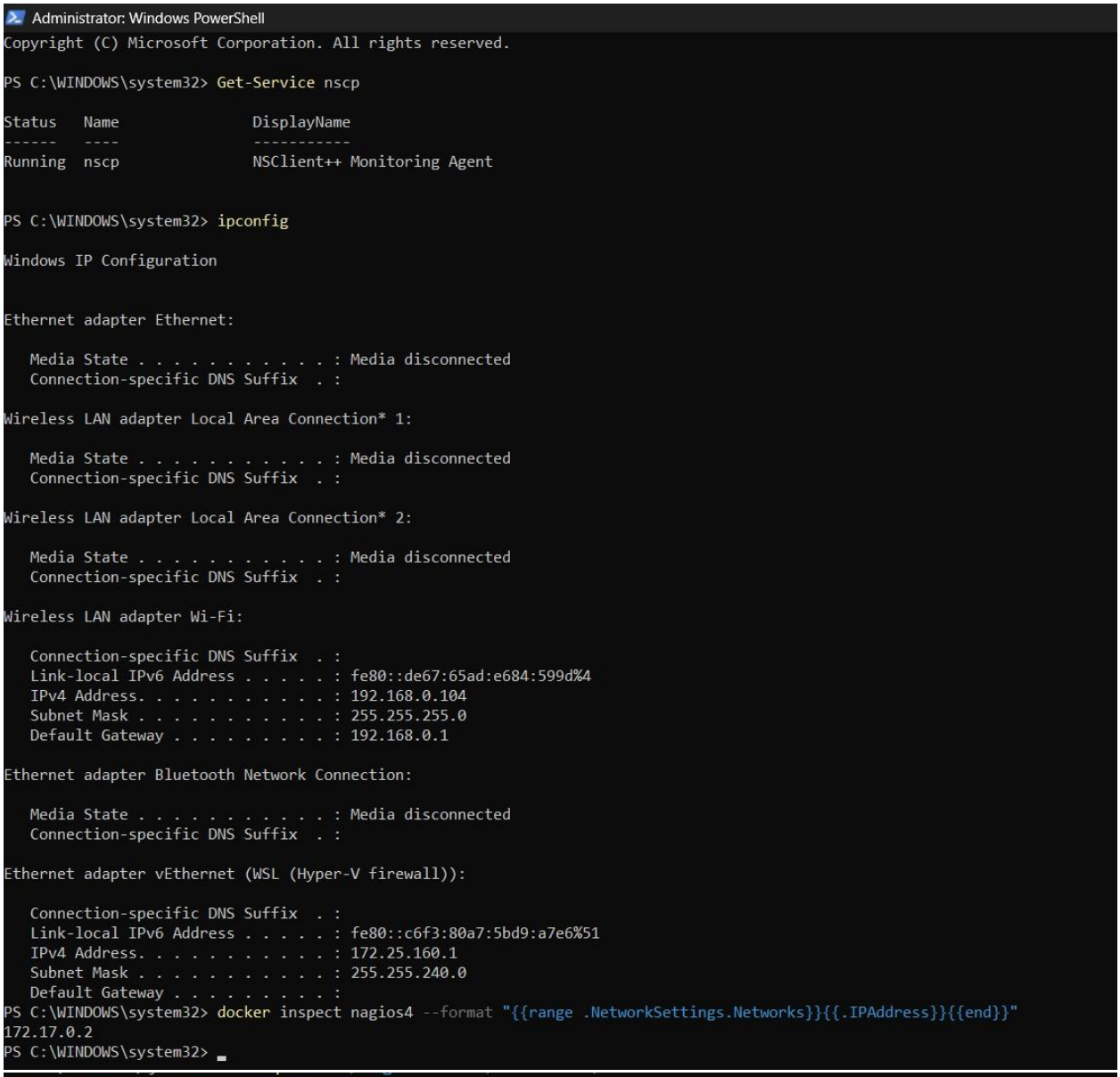

Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Neeraj>docker --version
Docker version 28.4.0, build d8eb465
C:\Users\Neeraj>wsl --version
WSL version: 2.6.1.0
Kernel version: 6.6.87.2-1
WSLg version: 1.0.66
MSRDC version: 1.2.6353
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26200.9278
C:\Users\Neeraj>wsl --status
Default Distribution: Ubuntu
Default Version: 2
C:\Users\Neeraj>


Command Prompt
C:\Users\Neeraj>docker pull jasonrivers/nagios:latest
latest: Pulling from jasonrivers/nagios
254354b70ac0: Pull complete
dc3a88012a3e: Pull complete
1413dee23aae: Pull complete
c5701b5e1540: Pull complete
a6b90b5bbb15: Pull complete
59215b8e47cf: Pull complete
cb259a83ac3d: Pull complete
e68fcda12c0c: Pull complete
54b9e993e315: Pull complete
f47c3af6bf44: Pull complete
fdbaa15b8986: Pull complete
f2f9a1b3ce48: Pull complete
0d59a81a5608: Pull complete
9288c4992679: Pull complete
e4dc1d4e010c: Pull complete
6fe99ef1bed3: Pull complete
2908452ff193: Pull complete
50ea270d34e9: Pull complete
1b9771f55a12: Pull complete
c42073aaf047: Pull complete
748b632e75e7: Pull complete
80b6ab7aef5d: Pull complete
37f75c8146e2: Pull complete
a7bd6dce6a77: Pull complete
7a7447982198: Pull complete
d003d4303177: Pull complete
7edde1931fd2: Pull complete
a03720c2c16b: Pull complete
a80db686d641: Pull complete
Digest: sha256:fee9cd6a2933e790dde613d70c3a603f96978d65fa6bbc3d617fce807e00f2e2
Status: Downloaded newer image for jasonrivers/nagios:latest
docker.io/jasonrivers/nagios:latest
C:\Users\Neeraj>docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest
f57433bcd9f3af746d209d5b3a19903ab32b0efde4a664daaf46fcff58c8a265
C:\Users\Neeraj>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
f57433bcd9f3 jasonrivers/nagios:latest "/usr/local/bin/star…" 13 seconds ago Up 11 seconds 0.0.0.0:8080->80/tcp, [::]:8080
->80/tcp nagios4

```
C:\Users\Neeraj>docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin

C:\Users\Neeraj>
```




Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\WINDOWS\system32> Get-Service nscp
Status Name DisplayName
Running nscp NSClient++ Monitoring Agent
PS C:\WINDOWS\system32> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::de67:65ad:e684:599d%4
IPv4 Address. . . . . : 192.168.0.104
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.0.1
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Ethernet adapter vEthernet (WSL (Hyper-V firewall)):
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::c6f3:80a7:5bd9:a7e6%51
IPv4 Address. . . . . : 172.25.160.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
PS C:\WINDOWS\system32> docker inspect nagios4 --format "{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}"
172.17.0.2
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Restart-Service nscp
PS C:\WINDOWS\system32> Get-Service nscp
Status Name DisplayName
Running nscp NSClient++ Monitoring Agent
PS C:\WINDOWS\system32> docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -4 -H 192.168.0.104 -2 -d 2 -L ADH
I (0.5.2.41 2018-04-26) seem to be doing fine...
PS C:\WINDOWS\system32> docker exec -it nagios4 ls -la /opt/nagios/etc
total 96